Tic-Tac-Toe

Android Mobile Game | Built with Java in Android Studio

Tic-Tac-Toe is a two-player mobile game developed for the Android platform using Java in Android Studio. The application was created as part of a mobile application development module, with the goal of practising the design and logic behind a simple but complete interactive game. The project covers several core areas of mobile app development, including user interface design, event handling for touch input, win and draw detection logic, and state management for restarting a game. The sections below walk through each part of the application, from the moment it opens to the final result screen.

App Icon

The application uses a custom-designed icon in place of the default Android Studio icon. This icon was chosen specifically to represent the theme of the game, so that anyone browsing their phone's app list can immediately recognise it as a Tic-Tac-Toe game without needing to open it first. Using a clear, purposeful icon is a small but important detail in mobile app design, as it directly affects how easily users can identify and return to an application.

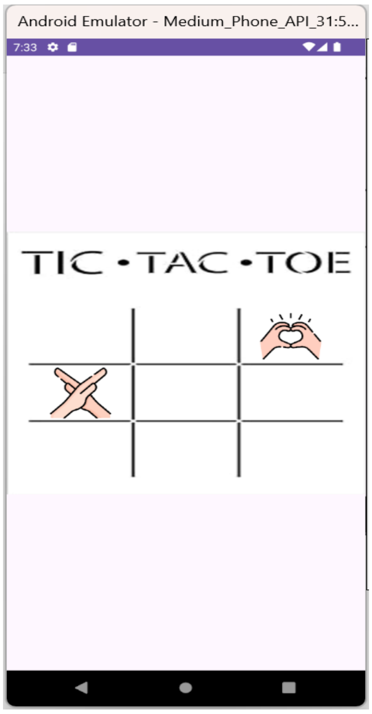
Opening Screen

This is the first screen a player sees after opening the application. The layout uses a simple white background paired with bold black grid lines and text, a deliberate design choice made to keep the game board clear and easy to read at a glance. Good contrast between the background and the game elements reduces visual strain and helps players quickly understand how to begin playing, which is especially important in a game meant to be picked up and played casually.

Player One Begins

Once the game board is displayed, the first player starts by tapping on any one of the nine available boxes. Each tap is registered through Android's touch event handling, which updates the selected box with the player's symbol. This interaction forms the foundation of the game loop, allowing the application to track which boxes have already been chosen and preventing the same box from being selected more than once.

Player Two Continues



After the first player has made their move, control passes to the second player, who can then choose from any of the eight remaining boxes. The application keeps track of which boxes are still open, ensuring that a box already claimed by one player cannot be selected again. This turn-based structure is central to how the game enforces fair play between the two participants.

Gameplay Continues

Play continues in this alternating pattern, with each player taking turns to fill the remaining empty boxes. Behind the scenes, the application continuously checks the board after every move to determine whether either player has formed a complete line of three matching symbols, whether horizontally, vertically, or diagonally. If no winning line is formed and all nine boxes become filled, the game recognises this as a draw rather than declaring a winner.

Draw Outcome

When all nine boxes have been filled without either player successfully completing a line of three, the game ends in a draw. The application displays a clear message informing both players of this outcome. From this screen, players are given the option to press the "Play Again" button, which resets the board completely and returns them to the opening screen shown in the first figure, ready for a new round.

Winning Outcomes

When either player successfully lines up three of their own symbols in a row, column, or diagonal, the game immediately detects this and ends the round in that player's favour. A result screen is shown to clearly announce the winner, whether it is the player using "O" or the player using "X". As with the draw outcome, the "Play Again" button is available on this screen as well, allowing both players to reset the board and begin a fresh match without needing to restart the entire application.

Self-Evaluation

Working on this project gave me my first hands-on introduction to building an interactive application in Android Studio. Through the process of designing the game board, handling player input, and implementing the logic to detect wins and draws, I gained a much clearer understanding of how mobile applications are structured and how different components work together to create a smooth user experience. This assignment also helped spark a stronger interest in mobile application development as a subject.

At the beginning of the module, I found the concepts fairly unfamiliar and challenging to apply. Completing this project marked a noticeable improvement compared to where I started, though I recognise there is still considerable room to grow in both my technical skills and my problem-solving approach going into future assignments. Along the way, I encountered several coding errors that required troubleshooting. I made use of online tutorials to work through some of these issues, and when that was not enough, I sought guidance from my lecturer, who helped me identify my mistakes and understand how to correct them.

Overall, this project was a valuable learning experience that strengthened both my technical foundation and my confidence in tackling unfamiliar problems independently. I intend to continue building on these skills through self-directed learning as I take on more advanced projects in the future.